```python

dp1 = [0] * n
dp2 = [0] * n

dp1[0] = entry[0] + line1[0]
dp2[0] = entry[1] + line2[0]

for i in range(1, n):
    dp1[i] = min(dp1[i-1] + line1[i],
                 dp2[i-1] + transfer2[i-1] + line1[i])

    dp2[i] = min(dp2[i-1] + line2[i],
                 dp1[i-1] + transfer1[i-1] + line2[i])

answer = min(dp1[n-1] + exit[0],
```

```
Minimum Production Time = 35

=== Code Execution Successful ===
```

```python

dp1 = [0] * n
dp2 = [0] * n

dp1[0] = entry[0] + line1[0]
dp2[0] = entry[1] + line2[0]

for i in range(1, n):
    dp1[i] = min(dp1[i-1] + line1[i],
                 dp2[i-1] + transfer2[i-1] + line1[i])

    dp2[i] = min(dp2[i-1] + line2[i],
                 dp1[i-1] + transfer1[i-1] + line2[i])

answer = min(dp1[n-1] + exit[0],
```

```
Minimum Production Time = 35

=== Code Execution Successful ===
```

```python

dp1 = [0] * n
dp2 = [0] * n

dp1[0] = entry[0] + line1[0]
dp2[0] = entry[1] + line2[0]

for i in range(1, n):
    dp1[i] = min(dp1[i-1] + line1[i],
                 dp2[i-1] + transfer2[i-1] + line1[i])

    dp2[i] = min(dp2[i-1] + line2[i],
                 dp1[i-1] + transfer1[i-1] + line2[i])

answer = min(dp1[n-1] + exit[0],
```

```
Minimum Production Time = 21

=== Code Execution Successful ===
```

```python

dp1 = [0] * n
dp2 = [0] * n

dp1[0] = entry[0] + line1[0]
dp2[0] = entry[1] + line2[0]

for i in range(1, n):
    dp1[i] = min(dp1[i-1] + line1[i],
                 dp2[i-1] + transfer2[i-1] + line1[i])

    dp2[i] = min(dp2[i-1] + line2[i],
                 dp1[i-1] + transfer1[i-1] + line2[i])

answer = min(dp1[n-1] + exit[0],
             dp2[n-1] + exit[1])
```

```
Minimum Production Time = 27

=== Code Execution Successful ===
```

```python

dp1 = [0] * n
dp2 = [0] * n

dp1[0] = entry[0] + line1[0]
dp2[0] = entry[1] + line2[0]

for i in range(1, n):
    dp1[i] = min(dp1[i-1] + line1[i],
                 dp2[i-1] + transfer2[i-1] + line1[i])

    dp2[i] = min(dp2[i-1] + line2[i],
                 dp1[i-1] + transfer1[i-1] + line2[i])

answer = min(dp1[n-1] + exit[0],
             dp2[n-1] + exit[1])
```

```
Minimum Production Time = 20

=== Code Execution Successful ===
```

```python

dp1 = [0] * n
dp2 = [0] * n

dp1[0] = entry[0] + line1[0]
dp2[0] = entry[1] + line2[0]

for i in range(1, n):
    dp1[i] = min(dp1[i-1] + line1[i],
                 dp2[i-1] + transfer2[i-1] + line1[i])

    dp2[i] = min(dp2[i-1] + line2[i],
                 dp1[i-1] + transfer1[i-1] + line2[i])

answer = min(dp1[n-1] + exit[0],
             dp2[n-1] + exit[1])
```

```
Minimum Production Time = 35

=== Code Execution Successful ===
```

```python
entry = [8, 10]
exit = [12, 15]

line1 = [5, 6, 4]
line2 = [7, 3, 5]

transfer1 = [2, 1]
transfer2 = [3, 2]

n = len(line1)

dp1 = [0] * n
dp2 = [0] * n

dp1[0] = entry[0] + line1[0]
dp2[0] = entry[1] + line2[0]
```

```
Minimum Production Time = 35

=== Code Execution Successful ===
```

```python

dp1 = [0] * n
dp2 = [0] * n

dp1[0] = entry[0] + line1[0]
dp2[0] = entry[1] + line2[0]

for i in range(1, n):
    dp1[i] = min(dp1[i-1] + line1[i],
                 dp2[i-1] + transfer2[i-1] + line1[i])

    dp2[i] = min(dp2[i-1] + line2[i],
                 dp1[i-1] + transfer1[i-1] + line2[i])

answer = min(dp1[n-1] + exit[0],
             dp2[n-1] + exit[1])
```

```
Minimum Production Time = 21

=== Code Execution Successful ===
```

```python

dp1 = [0] * n
dp2 = [0] * n

dp1[0] = entry[0] + line1[0]
dp2[0] = entry[1] + line2[0]

for i in range(1, n):
    dp1[i] = min(dp1[i-1] + line1[i],
                 dp2[i-1] + transfer2[i-1] + line1[i])

    dp2[i] = min(dp2[i-1] + line2[i],
                 dp1[i-1] + transfer1[i-1] + line2[i])

answer = min(dp1[n-1] + exit[0],
             dp2[n-1] + exit[1])
```

```
Minimum Production Time = 27

=== Code Execution Successful ===
```

```python
dp1 = [0] * n
dp2 = [0] * n

dp1[0] = entry[0] + line1[0]
dp2[0] = entry[1] + line2[0]

for i in range(1, n):
    dp1[i] = min(dp1[i-1] + line1[i],
                 dp2[i-1] + transfer2[i-1] + line1[i])

    dp2[i] = min(dp2[i-1] + line2[i],
                 dp1[i-1] + transfer1[i-1] + line2[i])

answer = min(dp1[n-1] + exit[0],
```

```
Minimum Production Time = 20

=== Code Execution Successful ===
```